ModelIt!

Student Activity Pack

How Do Doctors See Inside You Without Cutting You Open?

Modeling MRI Physics and Medical Imaging Technology

Grade 10L2 | NGSS HS-PS2-4, HS-PS4-4 | Structure and Function

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you know about how MRI machines create images of the inside of the body?

2 Why might a doctor choose MRI over an X-ray for certain types of injuries or conditions?

3 Draw a diagram showing what you think happens inside an MRI machine when it scans a patient.

4 What role do magnets play in medical imaging — how could a magnet help see inside a person?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Magnetic Field Strength | Coil Temperature | Image Resolution | Tissue Contrast | Patient Safety | Energy Consumption | Diagnostic Accuracy

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Standard 1.5T

Clinical standard field strength

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: High-Field 3T

Double the standard field strength

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Coil Failure

Superconducting coil warms above critical temperature

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Gradient Coil Precision — The accuracy of the additional magnetic field gradients that encode spatial location in the MRI signal — more precise gradients produce higher resolution images but generate louder noise and more heat

Scan Duration — The total time required to acquire sufficient data for a diagnostic-quality image — longer scans produce better images but increase patient discomfort, motion artifacts, and system operating costs

RF Pulse Sequence Design — The specific pattern and timing of radiofrequency pulses used to generate tissue-specific contrast — different sequences (T1-weighted, T2-weighted, FLAIR) highlight different tissue properties for different diagnostic purposes

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How can a machine create detailed images of your brain, muscles, and organs using nothing but a magnetic field and radio waves — without a single X-ray or incision? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate why MRI is preferred over CT scans for soft tissue imaging despite being more expensive and slower?**

2. **What does the coil failure scenario reveal about the engineering fragility of the superconducting magnet system?**

3. **Why is there an optimal Magnetic Field Strength rather than simply using the strongest possible magnet?**

4. **How does your model help explain the extreme cost of MRI machines ($1-3 million) and MRI scans ($500-$5,000)?**

5. **What are the limitations of modeling MRI physics with only seven components when real MRI involves quantum spin physics, Fourier transforms, and complex pulse sequence programming?**

8 STEM Challenge: Design a Next-Generation Portable MRI System

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YOUR ENGINEERING MISSION

Design a compact, lower-cost MRI system that could be deployed in rural clinics, ambulances, or developing countries where traditional MRI is unavailable.

THE SCENARIO

A medical technology startup wants to bring MRI capabilities to the 70% of the world's population that currently has no access to MRI imaging. Your team has been hired to design a portable MRI system that sacrifices some image quality for dramatic reductions in cost, size, and power requirements.

Engineering Design Thinking

1. WHAT TYPE OF MAGNET TECHNOLOGY WOULD YOU USE — SUPERCONDUCTING, PERMANENT, OR RESISTIVE ELECTROMAGNETS?
2. HOW WOULD YOU POWER THE SYSTEM IN LOCATIONS WITHOUT RELIABLE ELECTRICITY?
3. WHAT IMAGING PROTOCOLS WOULD YOU PRIORITIZE FOR THE MOST IMPACTFUL DIAGNOSES IN RESOURCE-LIMITED SETTINGS?
4. HOW WOULD YOU ADDRESS PATIENT SAFETY WITH A SYSTEM THAT MIGHT BE OPERATED BY NON-SPECIALIST TECHNICIANS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

MRI Physicists design and optimize MRI scanning protocols and develop new imaging techniques. They work in hospitals, research universities, and medical device companies, earning $90,000–$180,000/year. Biomedical Engineers who design medical imaging equipment earn $75,000–$150,000/year.